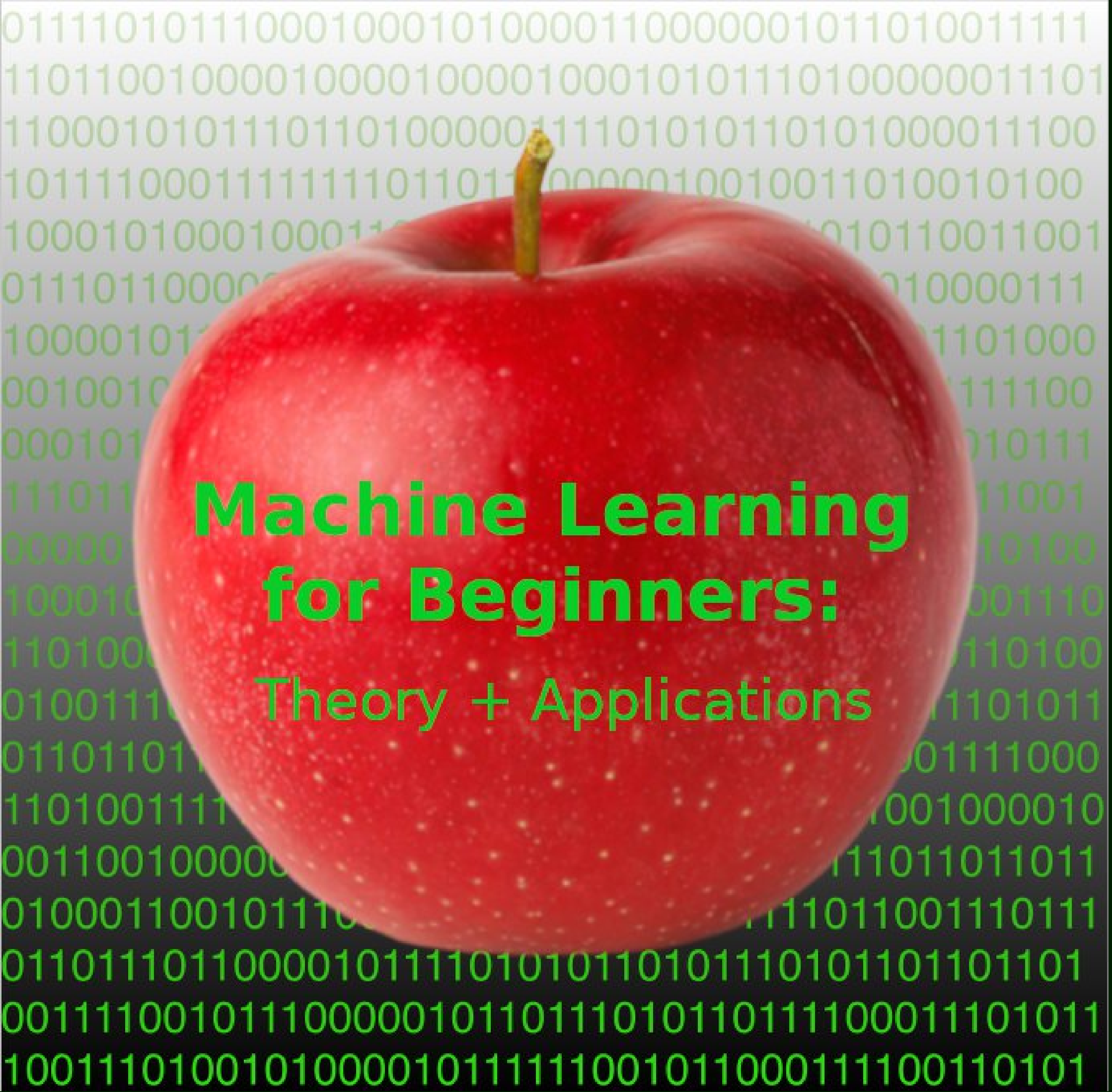


University of Bayreuth
Chair for Polymer Engineering
Prof. Dr. Holger Ruckdäschel

Title:

"Machine Learning for Beginners: Theory and Applications"

(Available in E-learning/Moodle)

Lecturer:

Dr. Rodrigo Q. Albuquerque

Lecture 8

Topics

Week 1: Introduction
ML types, data, ethics, case studies

Week 2: Preprocessing & *k*NN
Preprocessing, *k*-nearest neighbors, distance metrics, hyperparameters, cross-validation

Week 3: Linear Regression
Types, loss function, (over/under)fitting bias/variance, L1/L2 regularization, metrics

Week 4: Classification
Logistic regression, SVM, classification evaluation metrics, confusion matrix one-hot/label encoding

Week 5: Random forests
Decision trees, random forests, classification and regression applications

Week 6: PCA / KMeans
Dimensionality reduction, clustering

Week 7: Extra exercises

Week 8: Simple ANNs
Feedforward networks, activation functions, basic training

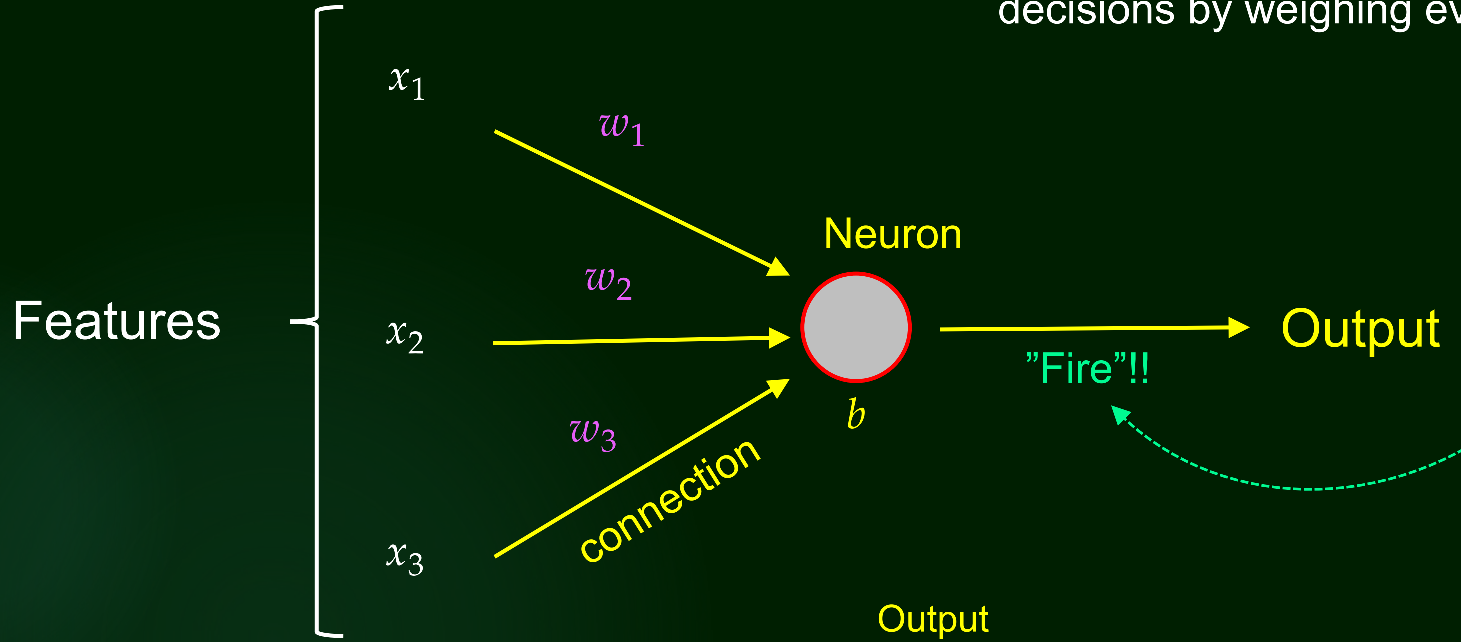
Week 9: Gaussian Processes
Kernels, gaussian processes

Week 10: Design of Experiments
DoE, random design, active learning, bayesian optimization

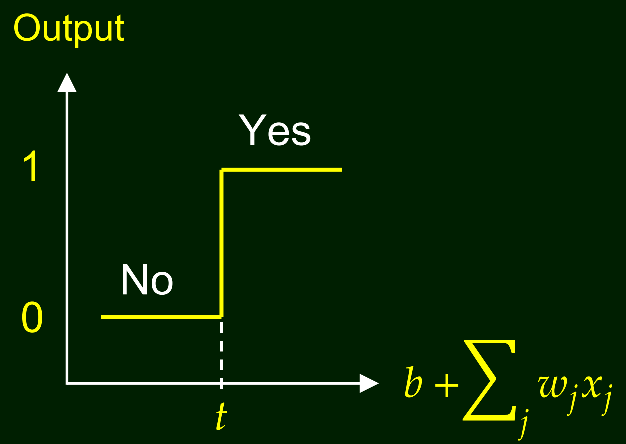
Weeks 11-12: Student projects

Perceptron

This is a kind of ”device” to make decisions by weighing evidence



$$\begin{cases} 0 & if \quad b + \sum_j w_j x_j \quad \leq \quad threshold\ t \\ 1 & if \quad b + \sum_j w_j x_j \quad > \quad threshold\ t \end{cases}$$



Curve is non-differentiable at the threshold value!
Derivative elsewhere is always zero
Training is not efficient!

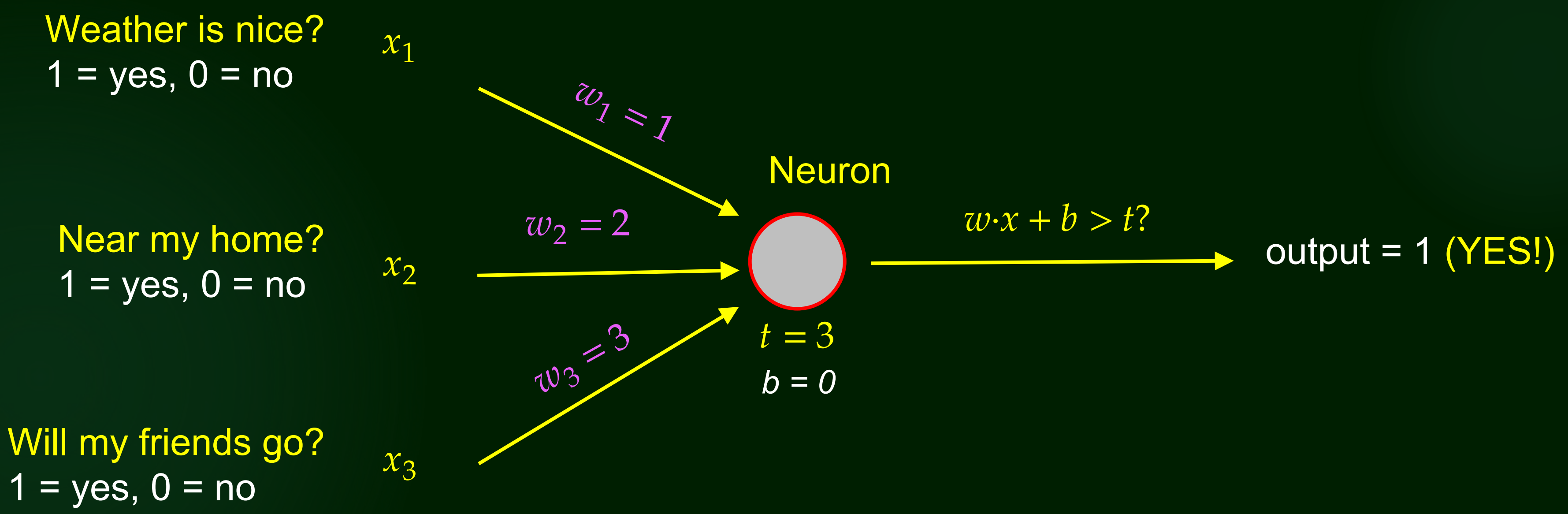
Frank Rosenblatt, 1957

Perceptron

Application

Should I go to the music festival on the weekend?

If output = 1 (YES), I go to the festival; if output = 0 (NO), I stay at home



$Therefore,\ w \cdot x + b > t$

$x_1 = 1\ (good\ weather),\ x_2 = 0\ (far\ away!),\ x_3 = 1\ (friends\ go) \Rightarrow w \cdot x + b = 1 \cdot 1 + 2 \cdot 0 + 3 \cdot 1 + 0 = 4$

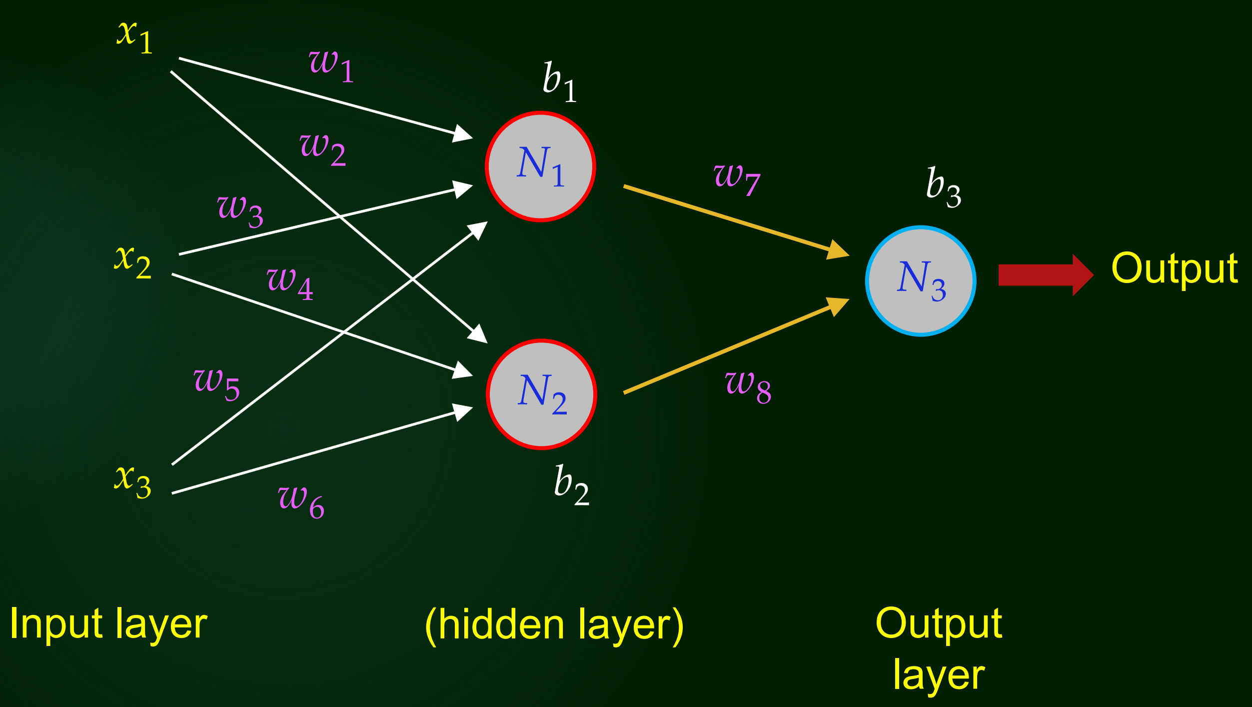
$x_1 = 1\ (good\ weather),\ x_2 = 1\ (nearby),\ x_3 = 0\ (friends\ don't\ go) \Rightarrow w \cdot x + b = 1 \cdot 1 + 2 \cdot 1 + 3 \cdot 0 + 0 = 3$

Perceptron

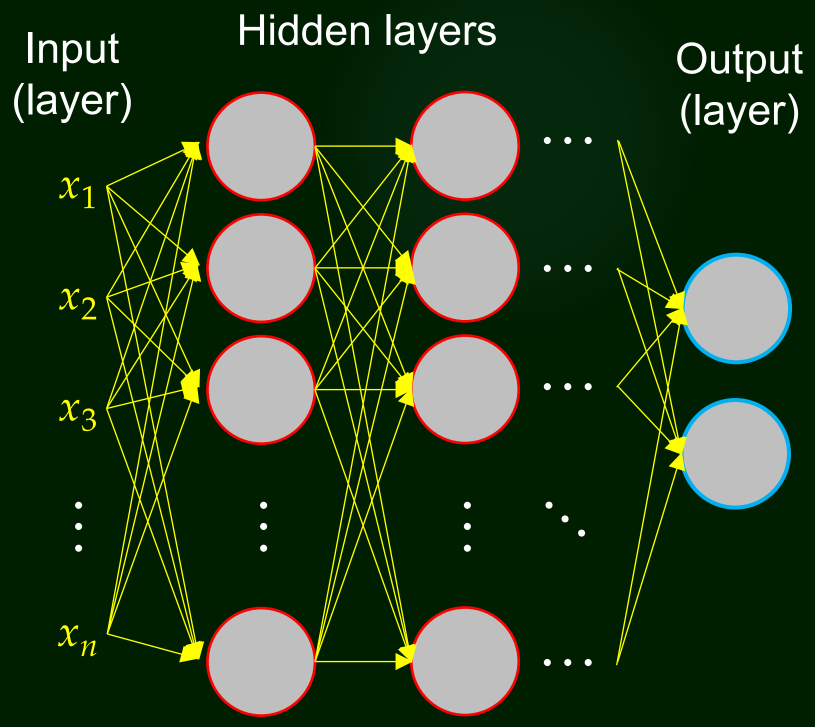
Multilayers

Simplification: threshold = 0

$$Output = \begin{cases} 0 & if \quad w \cdot x + bias \leq 0 \\ 1 & if \quad w \cdot x + bias > 0 \end{cases}$$



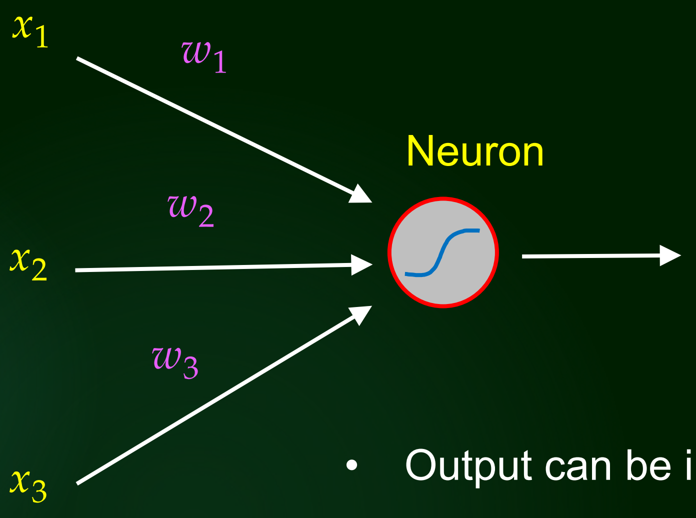
Problem: training VERY inefficient!



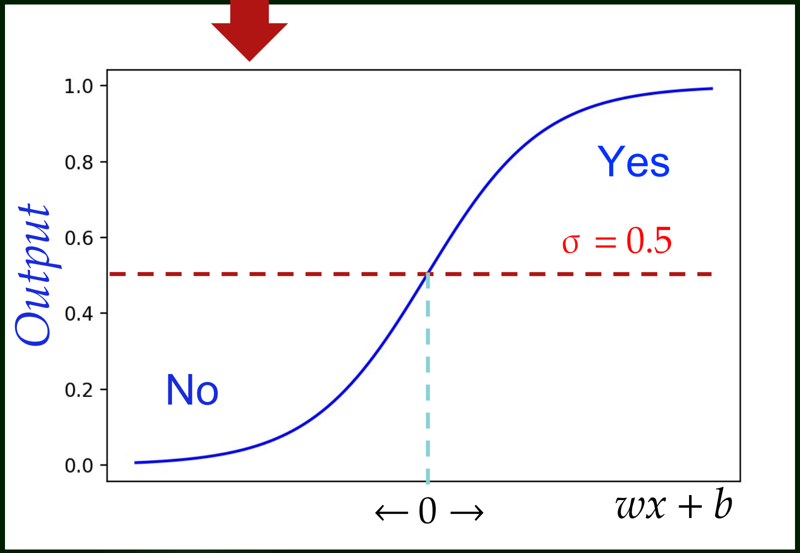
Make sofisticated decisions!

Artificial neural networks (ANNs)

Sigmoid neuron



$$\text{Output} = \sigma(wx + b) = \frac{\exp(wx + b)}{1 + \exp(wx + b)}$$



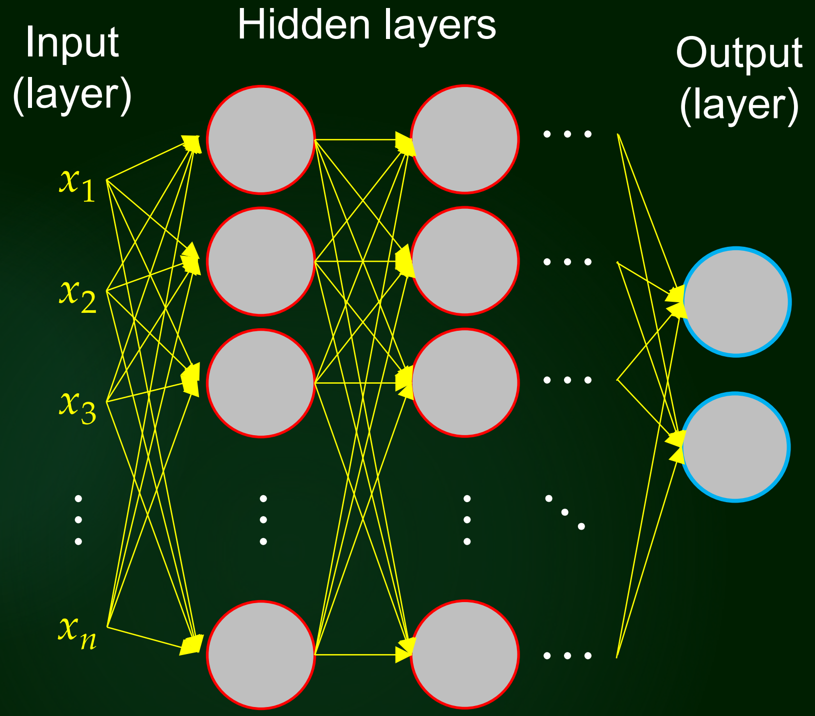
- Output can be interpreted as probability [0,1]
- Curve is differentiable (back-propagation possible!)
- Sigmoid = ”activation function”

Other activation functions:

Rectified linear unit ”ReLu”: *output = max(0,* $wx + b$*)*
Hyperbolic tangent ”Tanh”: *output = tanh(*$wx + b$*)*

ANNs

General concepts



Activation functions: σ, Relu, Tanh, etc.

Feed-forward NN's

Output from one layer = input to the next layer

Back-propagation

Use of $\partial\mathscr{L}/\partial w$ and $\partial\mathscr{L}/\partial b$ to train the model.

$$\Delta\text{output} \approx \sum_j \frac{\partial\,\text{output}}{\partial w_j}\Delta w_j + \frac{\partial\,\text{output}}{\partial b}\Delta b$$

Deep Learning

"Deep NN": Two or more hidden layers

Other ANN types

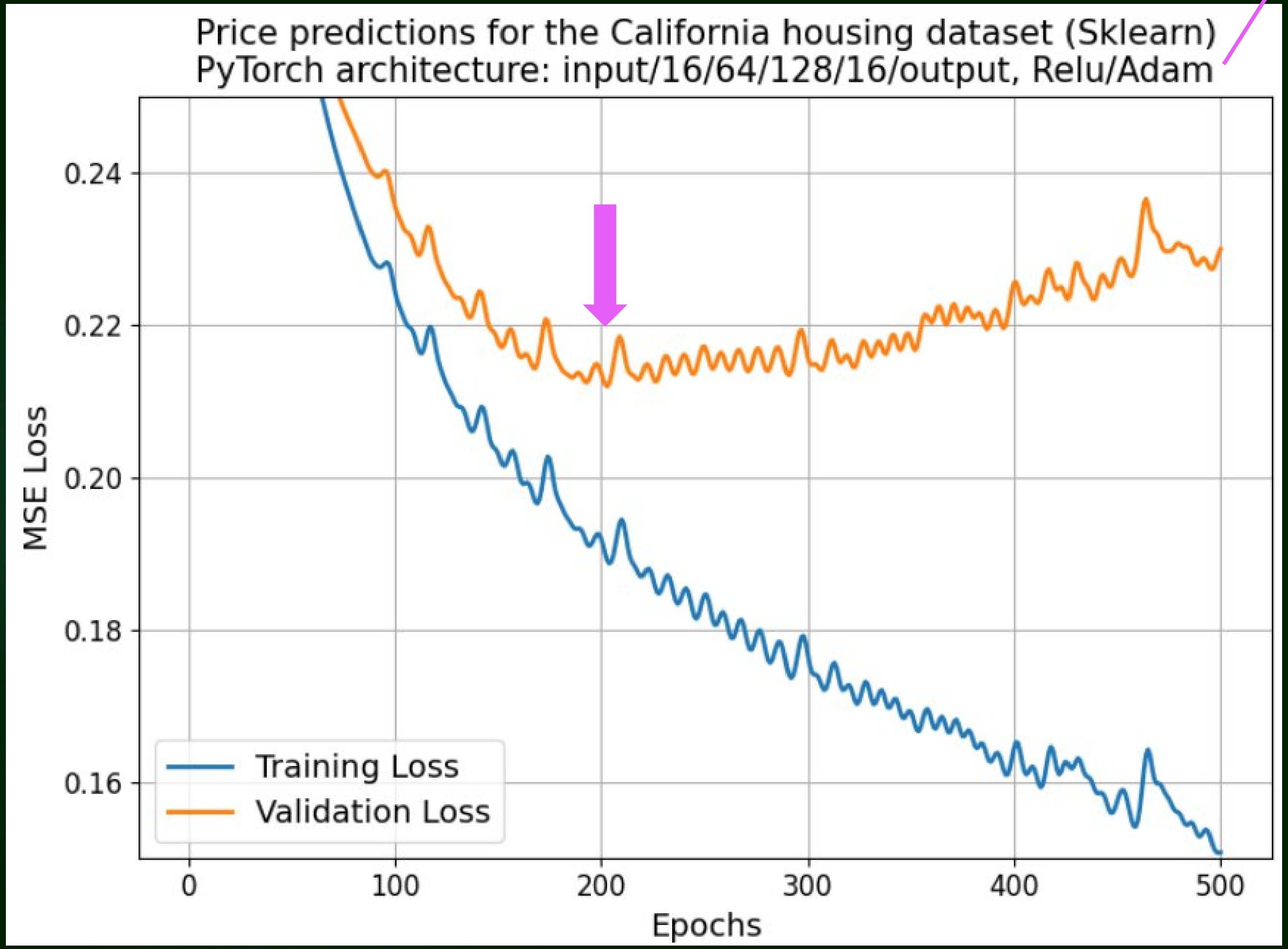
RNN, CNN, Autoencoder, GAN, Transformer, Ensemble NN

"Multilayer perceptron" (MLP) became a synonym of feed-forward ANNs, which use differentiable activation functions and are trained via back-propagation and gradient descent techniques.

ANNs

Training

$$w_{new} = w_{old} - \eta \cdot adjusted\ gradient$$



20% for validation, 80% (16512 samples) for training

Epoch

Training performed using the whole training data one single time

Mini-Batch Gradient Descent

Some samples (e.g., 32) are processed together to update w's and b's

Overfitting

ANN becomes too specialized on the training set after ca 200 epochs!

Early stopping

Stop training the ANN if validation loss didn't improve after XX epochs (e.g., XX = 5)

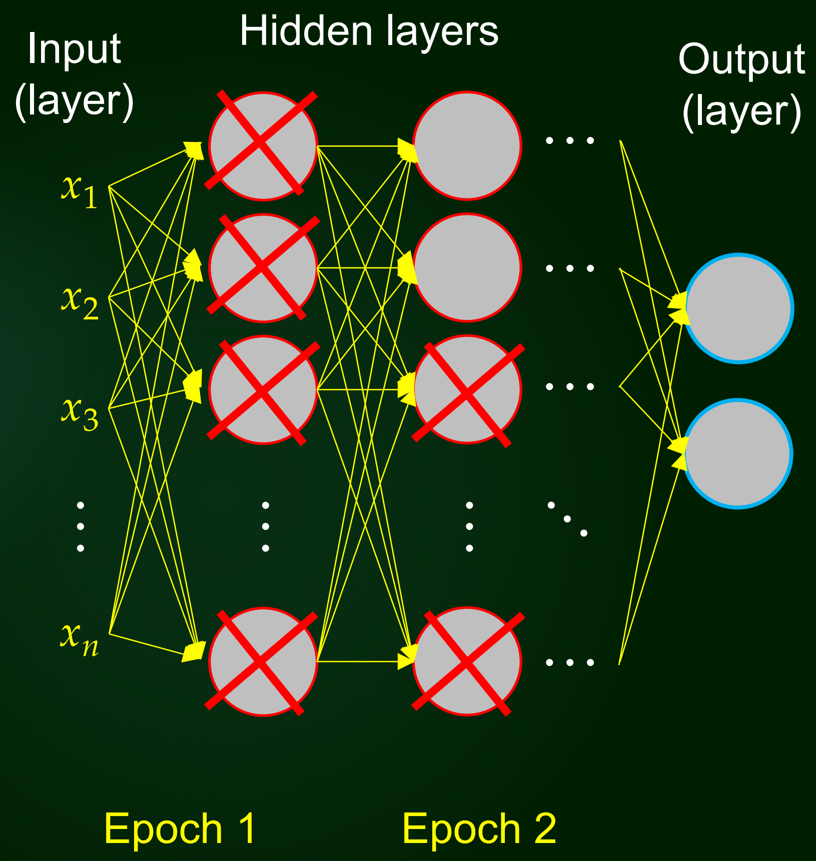
Regularization (L1 or L2)

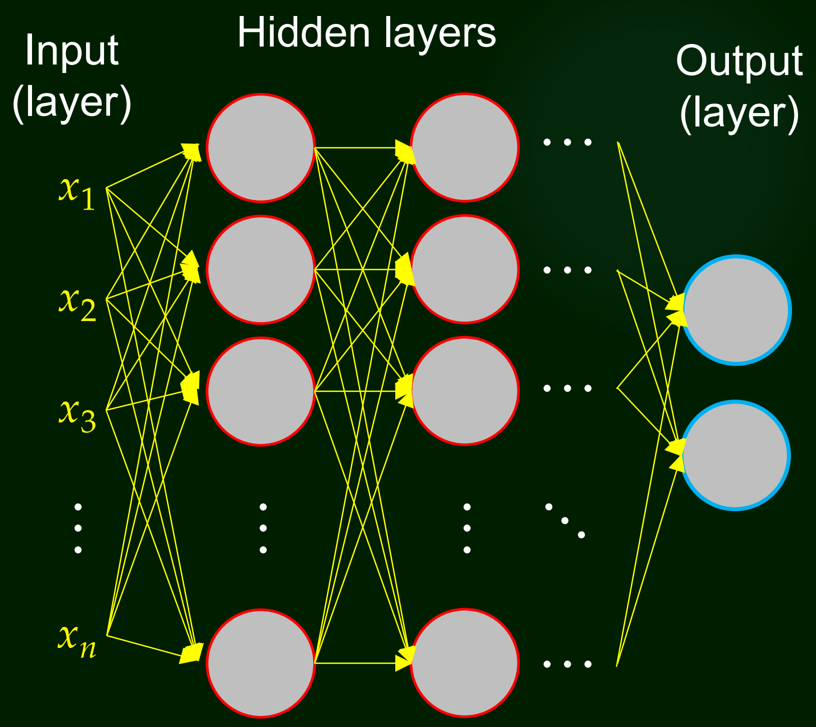
Apply penalties to large weight coefficients of either all hidden layers or selected ones

ANNs

Dropout prevents overfitting

Randomly selected neurons have output = 0 during training





During inference: all neurons are used

Exercises

1) Download the exercise ‘files8.zip’ from rqawork.github.io/teach
   and unzip it locally on your computer
2) Go to google colab (online) **or** launch jupyter notebook locally (offline)
3) Open/upload ‘exercise8.ipynb’ and wait for more instructions